

# Kiem Pham

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## Data Engineer - VNG Corporation

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 Ho Chi Minh City

 [dataspi.github.io/DataSpi](https://dataspi.github.io/DataSpi)

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Data/Analytics Engineer with 3+ years of experience turning raw operational data into trusted, well-modeled datasets for analytics. Strong in **Python**, **SQL**, **PostgreSQL**, **BigQuery**, **dbt**, and **AWS Glue**, with hands-on delivery of dimensional data models, tested ELT pipelines, and a self-service semantic layer (**Malloy**). Proven ability to bridge engineering and business stakeholders — having worked as both a BI/HR analyst and a data engineer — to deliver data products people actually trust and use, and to mentor non-technical teammates into full ownership of the systems I build.

## Highlighted Projects

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More details on my project, visit [Projects](#)

### 1. Real Estate Analytics Platform

*Sep 2025 – Present*

Built and now operate an end-to-end analytics engineering pipeline for Vietnam's real estate market (~200K+ for-sale & rental listings across HCMC & Hanoi), following a medallion architecture on **BigQuery**: async scrapers land raw listings, **dbt** staging models deduplicate and clean into a silver layer, and **dbt** mart models build a gold layer surfaced through a **Malloy** semantic layer for self-service metrics. Data quality is enforced with **dbt** schema tests plus custom singular tests (e.g. negative price/area, price-per-m<sup>2</sup> outliers). The pipeline runs on a scheduled, fail-fast orchestrator with logging, and results are published via an interactive **Looker Studio** dashboard and detailed analysis reports.

**Tech stack:** **Python**, **BigQuery**, **dbt**, **Malloy**, **curl\_cffi**, **BeautifulSoup**, **Google Looker Studio**, [GitHub](#), [Publication](#)

### 2. CI/CD-Orchestrated Logistics Reporting Pipeline

*Nov 2025 – Present*

Built an event-driven ELT pipeline on **GitHub Actions** for two freight-forwarding clients (retail & food logistics), replacing manual daily reporting. Scheduled workflows scrape order/booking data into **BigQuery** bronze tables, then automatically trigger a chained workflow (**workflow\_run**) that runs **dbt**

**build** for silver/mart models, followed by a report workflow that pushes milestone and TEU reports to stakeholders — a full scrape-to-report pipeline with zero manual intervention. Engineered a reusable composite action for secrets/env provisioning across pipelines, plus failure-artifact capture (**upload-artifact**) and manual **workflow\_dispatch** triggers for on-demand reruns and debugging.

**Tech stack:** **Python**, **GitHub Actions** (CI/CD, cron & event-driven orchestration), **dbt**, **BigQuery**, **Google Sheets API** **GitHub**

### 3. Building an Internal AI Assistant (Sep 2023 - Present)

Internal assistant for policy and knowledge lookup, letting employees retrieve company information in natural language. **OpenAI API**, **Langchain**, **Pinecone** - **GitHub**

### 4. Labor Market Analysis from Job Listing Sites (Aug 2023 - Present)

Automated job posting collection and analysis to track hiring trends in Vietnam's data job market. **Python**, **Selenium**, **BeautifulSoup** - **GitHub**, **Publication**

### 5. Employee Mental Health & Stress Survey Analytics (Aug 2023 - Sep 2023)

Analyzed employee and manager survey data to compare stress and mental health perceptions across cohorts. **Pandas**, **Google Forms** — **Project Link**



## Work experience

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### VNG Corporation - Data Engineer

Sep 2025- Now (1yr)

**Technical Core:** **Python**, **PostgreSQL**, **dbt**, **MetricFlow**, **AWS Glue**, **Hasura**, **Lightdash**.

#### Key Projects & Impact:

- Engineering Productivity Data Platform (TPMF): Designed a Kimball star-schema warehouse (a shared KPI-snapshot fact plus atomic-grain detail facts for drill-down) covering 45 engineering KPIs across three orgs (software delivery, QA/testing, People/HR), implemented as a **dbt** project on **PostgreSQL** with staging → intermediate → gold (mart) layers and KPI logic centralized in a **dbt** Semantic Layer (**MetricFlow**, incl. ratio/derived metrics). Built **Python** ETL for both file-based sources (Excel/CSV loaders into a raw landing schema) and live API sources (**GitLab**, **GitHub**, **SonarCloud**, an internal OAuth2 Service Desk API), with **dbt** schema and custom singular data-quality tests, running as cron pipelines on dedicated internal servers.
- Surfaced the modeled metrics through a read-only **MCP** server for natural-language querying in Claude, a **Lightdash** BI layer reading the marts directly, and an automated monthly executive report (self-contained offline HTML + **PPTX**).
- Data Platform & Migration: Designed and operated an **Odoo-to-ATS** data migration pipeline with reconciliation controls to maintain data integrity and prevent data loss at go-live.
- ETL/ELT Automation: Developed **Python** + **Boto3** automation for **AWS Glue** jobs, reducing manual operations and improving pipeline stability and run-time consistency.

- Managed **PostgreSQL/Hasura** workloads, including automated monthly partitioning, custom SQL functions, and scheduled maintenance (**pg\_cron**) to keep query latency stable as data volume grew.
- Data Modeling & Consumption: Integrated **Malloy** semantic modeling into the **IPay** environment to improve self-service analytics and natural language querying accuracy.
- Monitoring & Documentation: Established technical documentation standards and implemented monitoring/alerting flows with operations teams to improve reliability and incident response.

## LEGO Manufacturing Vietnam - Data Analyst

Sep 2024 - Aug 2025 (1yr)

- Reverse-engineered **UKG** timesheet data to resolve a high-stakes overtime calculation conflict (legal 300h/year cap vs. LEGO's actual pay policy vs. shift-based break rules); presented the logic to C&B leadership and cross-functional DA teams (Packing, Moulding), and it was adopted as the standard OT calculation used factory-wide — ahead of a global system fix that remained pending over a year later.
- Mentored a non-technical successor to full ownership of the PPC HR **Power BI** reporting system through a 3-week hands-on training program and detailed documentation; the system ran uninterrupted for 6+ months post-handover, and I remained the trusted technical advisor for two successive report owners after leaving LEGO.
- Designed and delivered 3 targeted reporting solutions for the PPC department, covering: headcount & workforce planning, turnover, D&I, L&D, overtime, absenteeism, retention policy analysis to support strategic decision-making.
- Produced monthly executive summaries from **Power BI** dashboards for senior leadership.
- Partnered with BI Manufacturing on Factory Huddle and Visual Factory reports used in daily operations and leadership meetings.
- Delivered an end-to-end data flow for LEGO Vietnam's 3-month Active Summer program using **Google Forms API + Python + Power BI**.
- Supported development of a **Power App** solution for gym consent and reservations, enhancing process efficiency and user experience.

## ITL Corporation

Data Analyst

Aug 2023 - Sep 2024 (1yr 1m)

- Conducted comprehensive analyses and generated detailed reports using data from the HRMS database.
- Developed a comprehensive data pipeline spanning database, data warehouse, and reporting phases using **MSSQL** and **PostgreSQL**.
- Streamlined data using **SQL** and **Python**.
- Visualized and created reports using **Power BI**.
- Collected and analyzed job listings from job-listing sites using **BeautifulSoup**, **Selenium**, and **Pandas** to generate comprehensive reports on the recruiting market.
- Led comprehensive internal surveys to assess key HR metrics, including stress levels, employee satisfaction, and employee engagement.

- Collaborated with IT to develop an internal AI assistant leveraging the [OpenAI API](#) and [Langchain](#).

## Human Resources Management Trainee

*Feb 2022 - Aug 2023 (1yr 7m)*

- Calculated monthly salary add-ons for the container fleet to ensure accurate and timely compensation adjustments.
- Provide support by answering questions related to company policies.
- Processed employee paperwork related to union matters and resignations.
- Planned employee costs for 2024.

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## Skills

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### Technical Skills

#### Analytics Engineering

- Dimensional Modeling (Star Schema), Medallion Architecture (Bronze/Silver/Gold), Data Warehouse & Data Marts
- ETL/ELT Pipelines, [dbt](#) (models, schema & custom data quality tests), Semantic Modeling ([Malloy](#))

#### Data & Programming

- Advanced [SQL](#) ([PostgreSQL](#), [BigQuery](#), [MSSQL](#))
- [Python](#) ([pandas](#), [BeautifulSoup](#), [Selenium](#), [matplotlib](#), [plotly](#))

#### Data Platform & CI/CD

- [PostgreSQL](#), [BigQuery](#), [dbt](#), [AWS Glue](#), [Airflow](#), [pg\\_cron](#), [Git](#), [Hasura](#)
- [GitHub Actions](#) (scheduled & event-driven workflow orchestration, dbt-in-CI, secrets management, reusable composite actions)

#### Business Analytics

- [Power BI](#), [DAX](#), KPI Development, Executive Reporting, Self-service Analytics

#### Collaboration

- Cross-functional stakeholder collaboration, technical documentation, mentoring & knowledge transfer, AI-assisted development ([Cursor](#), [VS Code](#))
- Public writing on data careers & self-taught learning — [Threads](#) post reached 4K+ views, 347 likes, 54 reposts; longer-form companion piece on [Substack](#)

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## Soft Skills

## Languages

- Vietnamese: Native or bilingual proficiency
  - English: Full professional proficiency (IELTS 7.0)
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## Education

### Bachelor Degree

- Human Resources Management - University of Economics HCMC - (*Sep 2018 - Dec 2021*)
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